

Aarul Dhawan

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EDUCATION

University of Illinois Urbana-Champaign

Masters in Computer Science (MCS) | Networks and Distributed Systems Concentration Aug 2025 – May 2026

Bachelors in Computer Science and Statistics (BS) | GPA: 3.9

Aug 2022 – Aug 2025

Relevant Coursework: Distributed Systems, Storage Systems, Database Systems, Communication Networks, Multimedia Systems

EXPERIENCE

Amazon Web Services

May 2025 – Aug 2025

Software Development Engineer Intern — S3 Express

Seattle, WA

- Re-architected a legacy Java load generator into a high-performance Rust system for S3 Express OneZone, improving throughput by **over 400%** and achieving consistent **sub-10ms p99 latency**.
- Added first-time support for benchmarking across multiple SDKs, enabling cross-language performance validation of S3 Express buckets in real-world customer scenarios.
- Deployed on AWS EKS with autoscaling to dynamically scale **hundreds of pods in under 60s**, while publishing **p50–p99.9 latency, error rates, and TPS metrics** into CloudWatch dashboards.

Epivara

Sept 2024 – Jan 2025

Software Engineer Intern

UIUC Research Park, IL

- Created an NLP-powered graph generation tool that interprets natural language queries to generate visualizations from experimental data stored in a SQL database.
- Led the establishment of the company's software department, building core systems, tooling, and engineering practices for AI-driven research workflows.

Epic Systems

May 2024 – Aug 2024

Software Developer Intern

Madison, WI

- Developed a cross-platform AI assistant using C#, Python, and React to compare echocardiogram studies, improving diagnostic efficiency in clinical settings.
- Estimated to save **\$3.4M+ annually** in labor costs for cardiologists across the U.S. healthcare system.
- Designed and implemented an LLM Evaluation Metric for patient summarization initiatives, and presented results to 200+ attendees while training 30+ Application Leads on adoption.

Disruption Lab, (Client: AMD)

Aug 2023 – May 2024

Software Engineer

Champaign, IL

- Modified data pipeline using dynamic mixing and noise augmentation techniques, increasing model robustness by 2.1x in noisy environments.
- Trained and fine-tuned ML models on AWS SageMaker across 200+ hours of audio data, reducing training time by 18% via pipeline optimizations.

PROJECTS

CS598 (Storage Systems): RDMA Distributed Key-Value Store

C, RDMA/RoCE

Distributed key-value store using **one-sided RDMA over RoCE** targeting **sub-millisecond latencies** vs S3 Express. Built with **Reed-Solomon erasure coding**, tiered RDMA/NVMe storage, and SWIM failure detection.

CS598 (GenAI Systems): Chunk-Level KV Cache Eviction

Python, vLLM, PyTorch

Fork of **vLLM v0.18.0** adding semantic chunk-level KV cache eviction for RAG workloads. Eviction scores combine retrieval relevance, attention signal, and recency; weights learned via an offline training pipeline.

CS 537: Bandwidth-Efficient Keyword Spotting

Python, PyTorch

Compared five transmission strategies for server-side keyword spotting using **PyTorch** CNNs and **Silero VAD** (raw PCM to learned embeddings), evaluating accuracy-bandwidth tradeoffs over a client-server socket architecture.

PayOff | Hack@Brown 2026

Swift, AWS

Peer-to-peer offline payment system using Apple Bonjour services — no WiFi or cellular required. Transactions use an **ECDH handshake** via Apple CryptoKit and execute automatically when any device reconnects.

ADDITIONAL INFORMATION

Languages: Rust, Java, Python, C++, C#, C, Swift, JavaScript, Haskell, R

Frameworks & Databases: React, Flask, Node.js, PostgreSQL, MySQL

Honors & Awards: Hack@Brown 2025 AWS Award Winner, UIUC Highest Distinction, UIUC Dean's List